2.5.1 Congruence Modulo

Two integers a and b are congruence modulo n if they differ by an integer multiple of n. That $a - b = kn$ for some integer k. This can also be written as $a \equiv b \pmod{n}$.

Here the number n is called modulus. In other words, $a \equiv b \pmod{n}$ means $a - b$ is divisible by n.

For example, $61 \equiv 5 \pmod{7}$ because $61 - 5 = 56$ is divisible by 7.

Note

- When a positive integer is divided by n, then the possible remainders are $0, 1, 2, \ldots, n-1$.
- Thus, when we work with modulo n, we replace all the numbers by their remainders upon division by n, given by $0,1,2,3,\ldots, n-1$.

Two illustrations are provided to understand modulo concept more clearly.

Illustration 1

To find $8 \pmod{4}$

With a modulus of 4 (since the possible remainders are 0, 1, 2, 3) we make a diagram like a clock with numbers 0,1,2,3. We start at 0 and go through 8 numbers in a clockwise sequence 1, 2, 3, 0, 1, 2, 3, 0. After doing so cyclically, we end at 0.

Therefore, $8 \equiv 0 \pmod{4}$

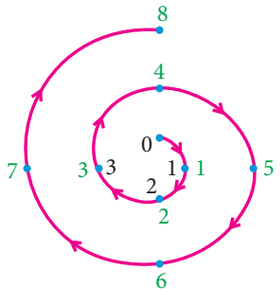


Fig.2.7

Illustration 2

To find $-5 \pmod{3}$

With a modulus of 3 (since the possible remainders are 0, 1, 2) we make a diagram like a clock with numbers 0, 1, 2.

We start at 0 and go through 5 numbers in anti-clockwise sequence 2, 1, 0, 2, 1. After doing so cyclically, we end at 1.

Therefore, $-5 \equiv 1 \pmod{3}$

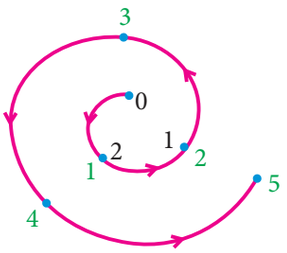


Fig.2.8

2.5.2 Connecting Euclid's Division lemma and Modular Arithmetic

Let m and n be integers, where m is positive. Then by Euclid's division lemma, we can write $n = mq + r$ where $0 \le r < m$ and q is an integer. Instead of writing $n = mq + r$ we can use the congruence notation in the following way.

We say that n is congruent to r modulo m, if $n = mq + r$ for some integer q.

$$n = mq + r$$
$$n - r = mq$$
$$n - r \equiv 0 \pmod{m}$$
$$n \equiv r \pmod{m}$$

Progress Check

1. Two integers a and b are congruent modulo n if _____.
2. The set of all positive integers which leave remainder 5 when divided by 7 are _____.

Thus the equation $n = mq + r$ through Euclid's Division lemma can also be written as $n \equiv r \pmod{m}$.

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